CSD208 Computer Architecture - Class Assignment 1

Instructions

- Assignment submitted after due date and time will not be evaluated and a score of zero will be awarded for this assignment.
- Since it's a class assignment, you are required to be present during the lecture timings, else your submission will not be considered for evaluation.

Due Date: 12:25 pm, Jan 10, 2020

Submitting this Assignment

You will submit (upload) this assignment on Blackboard. Email submissions will not be accepted.

- Write your answer after the given question in this document.
- Name this document as **CA1_CA2020_John_Doe.pdf** in case your name is John Doe.
- Upload a **pdf version** of the document.

Grading: This assignment has 2 points (with **weightage of 2%** in your overall 100 points)

Answer the following question in about 700-800 words.

1. **Comment on the architectural similarities between IBM System /360 and IBM z/Architecture. Also, highlight the differences in computer organizational features of these two architectures. Attach a list of references.**

Introduction

Few products in American history have had the massive impact of the IBM System/360—on technology, on the way the world works or on the organization that created them.
The architecture of the newly announced IBM System/360 featured these innovations:

1. An approach to storage which permits and exploits very large capacities, hierarchies of speeds, read-only storage for microprogram control, flexible storage protection, and simple program relocation.

2. An input/output system offering new degrees of concurrent operation, compatible channel operation, data rates approaching 5,000,000 characters/second, integrated design of hardware and software, a new low-cost, multiple-channel package sharing main-frame hardware, new provisions for device status information, and a standard channel interface between central processing unit and input/output devices.

3. A truly general-purpose machine organization offering new supervisory facilities, powerful logical processing operations, and a wide variety of data formats.

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The IBM System/360 (S/360) could be a family of mainframe systems that was declared by IBM on Apr seven, 1964, and delivered between 1965 and 1978. It had been the primary family of computers designed to hide the whole vary of applications, from little to giant, each business and scientific. The look created a transparent distinction between design and implementation, permitting IBM to unharness a set of compatible styles at completely different costs.

Z/Architecture known as ESA Modal Extensions (ESAME), is IBM's 64-bit instruction set design enforced by its mainframe computers. IBM introduced its 1st z/Architecture-based system, the z900, in late 2000. Later z/Architecture systems embrace the IBM z800, z990, z890, System z9, System z10, zEnterprise 196, zEnterprise 114, zEC12, zBC12, z13, and z14. Z/Architecture retains backward compatibility with previous 32-bit-data/31-bit-addressing design ESA/390 and its predecessors all the manner back to the 32-bit-data/24-bit-addressing System/360. However, all 24-bit and 31-bit problem-state application programs originally written to run on the ESA/390 design are going to be unaffected by this modification.

Each z/OS address house, known as a 64-bit address house, is sixteen exabytes in size. A z/OS address house is eight billion times the scale of the previous 2-gigabyte address house.
Each z/OS address space, called a 64-bit address space, is 16 exabytes in size. A z/OS address space is 8 billion times the size of the former 2-gigabyte address space.

Architectural similarities between IBM System /360 and IBM z/Architecture

Since the Z/Architecture is backward compatible with System/360. They retain most of the architectural similarities. The following are the similarities in the two architectures.

- Big endian byte ordering
- A processor with
  - 16 32-bit general purpose registers (R0-R15)
  - A 64-bit program status word (PSW), which describes (among other things)
    - Interrupt masks
    - Privilege states
    - A condition code
    - A 24-bit instruction address
  - An interruption mechanism, maskable and unmaskable interruption classes and subclasses
  - An instruction set. Each instruction is wholly described and also defines the conditions under which an exception is recognized in the form of program interruption.
- Manual control operations that allow
  - A bootstrap process (a process called Initial Program Load or IPL)
  - Operator-initiated interrupts
  - Resetting the system
  - Basic debugging facilities
  - Manual display and modifications of the system's state (memory and processor)

- An Input/Output mechanism - which does not describe the devices themselves

Some of the optional features are:

- Binary-coded decimal instructions
- Floating point instructions
- Timing facilities (interval timer)
- Key-controlled memory protection

Differences in computer organizational features of these two architectures

The two systems differ in multiple places. The way multiple features are implemented differs in the two:-

Customarily IBM Mainframe memory has been byte-addressable. This sort of memory is named "Focal Storage" and was utilized in System/360. IBM Mainframe processors through a great part of the 1980s and 1990s upheld another sort of memory: Expanded Storage. The Expanded Storage framework followed by z/Architecture is 4KB-page addressable. At the point when an application needs to get to information in Expanded Storage it should initially be moved into Central Storage. So also, information development from Central Storage to Expanded Storage is done in products of 4KB pages. At first page development was performed utilizing moderately costly directions, by paging subsystem code. The z/VSE Version 4, z/TPF Version 1 and z/VM Version 5 operating systems, and presumably their successors, require z/Architecture.

z/Architecture bolsters running various simultaneous working frameworks and applications regardless of whether they utilize diverse location sizes. This enables programming designers to pick the location size that is generally worthwhile for their applications and information structures. Stage Solutions Inc. (PSI) recently showcased Itanium-based servers which were good with z/Architecture. Hitachi centralized computers running more current arrivals of the VOS3 working framework actualize ESA/390 or more Hitachi-exceptional CPU directions, including a couple 64-piece guidelines. While Hitachi was likely motivated by z/Architecture, and officially teamed up with IBM on the z900-G2/z800 CPUs presented in 2002, Hitachi's machines are not z/Architecture-perfect.

On July 7, 2009, IBM once in a while of declaring another form of one of its working frameworks certainly expressed that Architecture Level Set 4 (ALS 4) exists, and is actualized on the System z10 and resulting machines. The ALS 4 is likewise determined in LOADxx as ARCHLVL 3, though the prior z900, z800, z990, z890, System z9 indicated ARCHLVL 2.

The basic System/360 models utilized the Basic Operating System/360 (BOS/360), Tape Operating System (TOS/360), or Disk Operating System/360 (DOS/360, which developed into DOS/VS, DOS/VSE, VSE/AF, VSE/SP, VSE/ESA, and afterwards z/VSE).

The bigger models utilized Operating System/360 (OS/360). IBM built up a few variants of OS/360, with progressively ground-breaking highlights: Primary Control Program (PCP),

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Multiprogramming with a Fixed number of Tasks (MFT), and Multiprogramming with a Variable number of Tasks (MVT).

IBM portrayed MVPG as "moves a solitary page and the focal processor can't execute some other guidelines until the page move is finished." The overhead of moving single and gatherings of pages among Central and Expanded Storage was decreased with the presentation of the MVPG (Move Page) guidance and the ADMF (Asynchronous Data Mover Facility) capacity. This was something missing in the previous System360 Architecture.

References:

1. Development and Attributes of z/Architecture, IBM Journal of Research and Development, 2002.
2. Columbus State Journal - http://csc.columbusstate.edu/woolbright/zarchitecture.pdf
3. "IBM System/360 Dates and Characteristics". IBM. 2003-01-23.
4. IEEE Annals of the History of Computing. IEEE Computer Society.
5. ACM Abstracts
6. "MVPG faster than MVCL for aligned pages?" IBM-MAIN.
7. Wikipedia.com - https://en.wikipedia.org/wiki/Z/Architecture